

Canonical ReLU Swing-by: certified conditional timing

The full-tube conditional timing theorem is closed. It separates the repository statistic’s sign transition from every full OOD-error minimizer on the certified time window, uniformly over an explicit sector. Exact initialization-to-entry reachability remains open. The exact fixed-pattern moment equations and their event-transfer limitation are stated separately.

1 Conditional canonical timing and minimizer localization

Use the exact binary canonical dataset, exact archived reference, and fixed 71-neuron ID-selected cohort of the existing continuation certificate. The loss is $\mathcal{L} = (2N)^{-1} \sum_n \|f_\theta(x_n) - x_n\|^2$, with $N = 4000$, $h = 200$. All times below are physical times. Set

$$r = \sin(\pi/36), \quad k = \cos(\pi/36), \quad \xi_\alpha = (\cos \alpha, \sin \alpha), \quad \mathcal{I} = \{\alpha : |\alpha + 17\pi/36| < 10^{-6}\}.$$

For the fixed ungated strong block $M_I = (m_{ab})$, retain exactly the repository statistic

$$Q_G = [(m_{11} - 1)r - m_{12}](r\dot{m}_{11} - \dot{m}_{12}) + r\dot{m}_{21} - \dot{m}_{22}.$$

The factors k are not inserted into this auxiliary statistic. Write $q_E(\alpha, t) = \frac{d}{dt} \frac{1}{2} \|f_t(\xi_\alpha) - \xi_\alpha\|^2$, with time derivatives understood almost everywhere.

Theorem 1.1 (Conditional sign separation and minimizer localization). *Every global Clarke flow satisfying the existing entry condition*

$$\|\theta(3.4) - \hat{\theta}_{3.4}\|_2 \leq 1.2 \cdot 10^{-5}$$

has the following properties. Put

$$[g_-, g_+] = [3.5738, 3.6348], \quad [e_-, e_+] = [3.6384, 3.7152].$$

For every compatible training-selector realization,

$$\begin{array}{llll} Q_G < 0 & \text{a.e. on } [3.4, g_-], & Q_G > 0 & \text{a.e. on } [g_+, 3.9], \\ q_E(\alpha, \cdot) < 0 & \text{a.e. on } [3.4, e_-], & q_E(\alpha, \cdot) > 0 & \text{a.e. on } [e_+, 3.9] \quad (\alpha \in \mathcal{I}). \end{array}$$

Consequently every minimizer of $E(\xi_\alpha, \cdot)$ on $[3.4, 3.9]$ lies in $[e_-, e_+]$, uniformly for $\alpha \in \mathcal{I}$. The sign-transition brackets are strictly separated:

$$e_- - g_+ = \frac{18}{5000} = 0.0036 > 0.$$

Throughout the interval $[g_+, e_-]$, almost everywhere and uniformly in $\alpha \in \mathcal{I}$,

$$Q_G > 1.21 \cdot 10^{-7}, \quad q_E(\alpha, \cdot) < -5.44 \cdot 10^{-7}, \quad q_E(\alpha, \cdot) - Q_G < -2.047 \cdot 10^{-4}.$$

For the central probe $\alpha = -17\pi/36$, the tighter minimizer bracket is $[3.6394, 3.7136]$, giving separation $23/5000 = 0.0046$. No uniqueness or continuity of a Q_G zero, unique OOD minimizer, or initialized reachability is asserted.

Proof. The existing flow certificate is used without modification. On each of its 2500 intervals $[t_j, t_{j+1}]$, its nonnegative scalar comparison is nondecreasing, so its terminal bound r_{j+1} bounds the distance to the cubic reference throughout that entire interval. This justifies using the recorded **radius_upper**, rather than the larger prescribed guard radius. No flow calculation is rerun.

Let y_j be the exact binary midpoint used by the timing verifier. If d_j bounds motion of the cubic reference about its interval midpoint, and ϵ_j bounds midpoint rounding, then

$$\|\theta(t) - y_j\| \leq R_j := r_{j+1} + d_j + \epsilon_j \quad (t_j \leq t \leq t_{j+1}).$$

The cubic derivative bounds and all these additions are outward-rounded. For $P_j = \|y_j\|$, $\bar{P}_j = P_j + R_j$, the static-neighborhood proposition in the continuation certificate provides a polynomial-field Lipschitz bound L_j and an actual-training-gate correction B_j . The latter encloses all compatible selectors, including every ambiguous training gate.

For either potential, put $g = \nabla\Phi$, $H \geq \sup \|Dg\|$ on this ball. The checked static inequality is

$$|g(\theta)^T F(\theta) - g(y_j)^T F(y_j)| \leq R_j \{H(\|F(y_j)\| + L_j R_j) + \|g(y_j)\| L_j\} + (\|g(y_j)\| + H R_j) B_j.$$

For Q_G , $\Phi = \frac{1}{2}[(m_{11} - 1)r - m_{12}]^2 + r m_{21} - m_{22}$. For the full error, use the actual probe-active matrix and the quadratic reconstruction potential. For the discrepancy, apply the same inequality *directly* to $g_E - g_G$, with $H_E + H_G$ as a valid derivative bound; separately enlarged rate intervals are not subtracted.

Probe signs are certified on each entire tube. For each reference preactivation q its four Bernstein control coefficients are

$$q_0, \quad q_0 + \frac{1}{3} h \dot{q}_0, \quad q_1 - \frac{1}{3} h \dot{q}_1, \quad q_1, \quad h = 1/5000.$$

Their interval convex hull bounds the exact reference cubic. Subtract the parameter-tube perturbation $r_{j+1} \|\xi\|$ and the sector cost $(\sup_t \|\hat{u}_i(t)\| + r_{j+1}) 10^{-6}$ from the oriented margin. The minimum resulting margin over all steps and neurons exceeds $1.40926 \cdot 10^{-5}$. Thus no probe gate is unresolved. The smooth probe potential may be extended polynomially to the larger midpoint ball even when only the reference tube is known to preserve its gates: equality with the actual error is required only on the actual tube.

On each tube the angular transport cost for the error rate is bounded by

$$2(1 + \bar{P}_j^2/2) \bar{P}_j \sup \|F\| 10^{-6}.$$

Indeed the probe-cell matrix has norm at most $\bar{P}_j^2/2$, its time derivative at most $\bar{P}_j \|F\|$, and differentiation of $\langle M\xi - \xi, \dot{M}\xi \rangle$ in ξ gives this Lipschitz bound. Q_G is independent of the varied probe angle, so the same cost applies to $q_E - Q_G$.

All 2500 interval enclosures pass. The negative-prefix and positive-suffix margins, rounded outward, are respectively

Quantity	negative prefix: $-$ upper	positive suffix: lower
Q_G	$> 1.0839 \cdot 10^{-7}$	$> 1.2158 \cdot 10^{-7}$
q_E , central probe	$> 3.2526 \cdot 10^{-7}$	$> 3.5073 \cdot 10^{-7}$
q_E , entire sector	$> 5.4476 \cdot 10^{-7}$	$> 1.9957 \cdot 10^{-8}$

The sign-unresolved steps are enclosed in exactly the stated brackets. The 18 whole steps between g_+ and e_- have both required strict signs. On these steps the direct central discrepancy enclosure is contained in $[-3.05302 \cdot 10^{-4}, -2.08690 \cdot 10^{-4}]$; after angular transport its upper bound is less than $-2.04714 \cdot 10^{-4}$. The exact step ledger, every sign-unresolved index, input hashes, and neighboring strict margins appear in `TIMING_FULL_TUBE_CERTIFICATE.csv` and its JSON summary.

The error is absolutely continuous and hence continuous on the compact interval. Integrating the strictly negative prefix bound excludes any minimizer before e_- , and integrating the positive suffix bound excludes any minimizer after e_+ . The Q_G sign bounds establish an earlier negative-to-positive sign transition bracket, without invoking the intermediate value theorem for a discontinuous training field. $\square$

Independent interior witness. The proposed $t = 3.64$ witness is independently reproduced. In the static ball of radius $2.6457445093 \cdot 10^{-5}$ (enclosed upward), outward-rounded bounds are

$$\begin{aligned} Q_G &\in [2.7405 \cdot 10^{-5}, 2.5834 \cdot 10^{-4}], \\ q_E &\in [-2.1933 \cdot 10^{-4}, -8.7767 \cdot 10^{-6}], \\ q_E - Q_G &\in [-3.0089 \cdot 10^{-4}, -2.1296 \cdot 10^{-4}]. \end{aligned}$$

The speed is below 0.330122389. The certified radius at 3.64 leaves more than $2 \cdot 10^{-6}$ available, while $6 \cdot 10^{-6}$ time units cost less than $1.980735 \cdot 10^{-6}$ in parameter distance. A first-exit argument gives $Q_G > 0$ and $q_E < 0$ on $[3.64, 3.640006]$ for the central probe. The displayed rounded intervals are deliberately outward.

Scope and provenance. The theorem concerns the supplied exact binary dataset and every entering Clarke flow; the archived exact initial weights have not been connected to this entry set. The historical archive has no independent raw-data hash. The supplied binary data match the pinned repository generator under the recorded runtime; this is not a high-probability theorem. Neither the empirically interpolated times nor a uniqueness assumption is used.

2 Exact dynamics of training-pattern moments

The residual convention matters. In this section define $\rho_n = f_\theta(x_n) - x_n$ and $A_g = N^{-1} \sum_n g_n \rho_n x_n^T$. If instead the residual is $x_n - f_\theta(x_n)$, all three displayed minus signs below become plus signs. Both conventions describe the same flow.

Lemma 2.1 (Fixed-pattern moment dynamics). *On an open time interval where every training preactivation is nonzero and the neuronwise training patterns are fixed, let*

$$I_g = \{i : \mathbf{1}_{u_i^T x_n > 0} = g_n \text{ for all } n\}, \quad M_g = \sum_{i \in I_g} w_i u_i^T, \quad U_g = \sum_{i \in I_g} u_i u_i^T, \quad W_g = \sum_{i \in I_g} w_i w_i^T.$$

Then the exact closed equations are

$$\begin{aligned} \dot{M}_g &= -A_g U_g - W_g A_g, \\ \dot{U}_g &= -A_g^T M_g - M_g^T A_g, \\ \dot{W}_g &= -A_g M_g^T - M_g A_g^T, \quad f(x_n) = \sum_g g_n M_g x_n. \end{aligned}$$

All moments include their entire groups; no concentration or low-rank approximation is assumed.

Proof. Direct differentiation of the stated normalized loss gives $\dot{w}_i = -A_g u_i$ and $\dot{u}_i = -A_g^T w_i$ for $i \in I_g$. Differentiate each outer product and sum. For example $\dot{w}_i u_i^T + w_i \dot{u}_i^T = -A_g u_i u_i^T - w_i w_i^T A_g$. The other two identities follow in the same way, with the transposes shown. The network identity follows from $(u_i^T x_n)_+ = g_n u_i^T x_n$. Consequently A_g depends only on the fixed patterns, the data, and the moments, proving closure on this interval. $\square$

Put

$$K_g = \begin{pmatrix} U_g & M_g^T \\ M_g & W_g \end{pmatrix}, \quad B_g = \begin{pmatrix} 0 & A_g^T \\ A_g & 0 \end{pmatrix}.$$

Then $K_g \succeq 0$, $\dot{K}_g = -B_g K_g - K_g B_g$ and $\text{tr}(U_g - W_g)$ is constant on the fixed-pattern interval. The same equations hold for a partition refined by fixed cohort membership and fixed probe signs, since every subgroup with training pattern g uses the same A_g .

What the quotient retains and loses. The training-pattern moments determine the training outputs on that cell. They do not, by themselves, determine a new probe's output, even if every individual probe sign is specified. For a concrete example use a single training vector e_1 , $a, b > 0$, and two balanced neurons $w_i = u_i$. Compare

$$u_1 = (a, b), \quad u_2 = (a, -b) \quad \text{with} \quad u'_1 = (7a/5, b/5), \quad u'_2 = (a/5, -7b/5).$$

This is an orthogonal mixing of the two neuron columns. Both training patterns are active, and in both states the $-e_2$ probe is inactive on neuron 1 and active on neuron 2. All three training-group moments are identical, equal to $\text{diag}(2a^2, 2b^2)$. Yet

$$f(-e_2) = (ab, -b^2), \quad f'(-e_2) = (7ab/25, -49b^2/25).$$

Thus probe-refined moments (or an equivalent signed probe observable) are necessary for an OOD interface. This example concerns the sufficiency of proposed observable data; it is not a counterexample to canonical initialized reversal.

Gate-event transfers. Let $y_i = (u_i, w_i)$ and $K_i = y_i y_i^T$. At an isolated switch of neuron i from training group g to g' , the underlying state is continuous, but its group representation changes by

$$K_g^+ = K_g^- - K_i, \quad K_{g'}^+ = K_{g'}^- + K_i.$$

Simultaneous switches sum their individual contributions. The group moment does not identify K_i . Indeed different rank-one decompositions, including the orthogonal mixing above, have the same K_g and different K_i . An estimate of the form

$$\|K_i - \hat{K}_i\| \leq C \|K_g - \hat{K}_g\| \tag{1}$$

therefore fails for every finite universal C . A transfer enclosure must retain the boundary neuron's rank-one moment or other sufficient geometry. This is an information loss of the unaugmented quotient, not a failure of the proposed augmented quotient/event strategy.

Reference census. The deterministic census of the stored numerical reference has 53 distinct training patterns at $t = 0$, 121 at $t = 1$, 140 at $t = 2$, 134 at $t = 3.4$, and 134 at $t = 4$. A 29-neuron inactive training group persists at these reference knots. Over 20000 adjacent reference-knot pairs there are 185211 sample–neuron sign differences. These are diagnostic counts, not certified actual-flow events; multiple crossings between knots could be missed. At entry a 16-neuron training-pattern group mixes strong-cohort membership. It must be split by membership to recover the repository Q_G from group moments. The stepwise CSV also records all group sizes and a nonrigorous linear-motion near-boundary count. No census number is a proof premise.